

ARK-490 — Authority Engine · Sustained Throughput

Status: PREREGISTRATION (locked before execution)

Experiment Date: 2026-07-18

Series: ExecutionProof P02 Latency/Throughput/Scale

Question

What is the sustained throughput — authority decisions/second maintained over an extended window — of the reference **Authority Engine**, and does it degrade relative to burst (ARK-489)?

Analogous to ARK-485 (Verification Decision sustained), applied to the current-state authority check.

Scope & Honesty Boundary

Minimal in-process reference `AuthorityEngine`, built to MEASURE, not the production engine. Claims bounded to this reference under the stated load.

Hypothesis

Sustained throughput $\geq 100,000$ decisions/second over 60 seconds at **100% accuracy** — lower than burst due to GC and memory pressure over the longer window.

Component Under Test

Reference `AuthorityEngine` (exact, fail-closed). `engine/perf_harness.py --dimension sustained`.

Methodology

- **Warmup:** 5 seconds (excluded)
- **Sustained window:** 60 seconds of continuous decisions on the valid grant path
- Count total and correct decisions; throughput = total / elapsed

Metrics

Primary: `sustained_throughput_dps`, `accuracy`

Secondary: `total_decisions`, `duration_sec`

Thresholds (Gate Conditions)

C1 (Throughput): $\geq 100,000$ dec/s

C2 (Accuracy): 100%

C3 (Correctness gate): valid→ALLOW, mutated→DENY, revoked→DENY

Verdict: PASS if $C1 \wedge C2 \wedge C3$; otherwise FAIL

Kill-Gate

K1: Sustained < 10% of burst (ARK-489) → possible implementation error.

K2: Accuracy < 100% → guard logic compromised → GATE-STOP.

Honest Findings Commitment

Any deviation (degradation patterns, accuracy failures) disclosed in RESULTS.md.

Execution Protocol

1. **Lock** prereg + MANIFEST.txt hashes
 2. **Execute** `engine/perf_harness.py --dimension sustained`
 3. **Record** to `results/sustained_results.json`
 4. **Evaluate** against C1–C3
 5. **Report** in RESULTS.md
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Preregistered: 2026-07-18

Investigator: Remnant Fieldworks Inc. / ExecutionProof Research Program

Covenant: RF Standing Covenant (preserve outcomes, bound claims, honest disclosure)